INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

## Degree: BCA, Stream: BCA

## Term – II, Second Semester Examination, March - 2025

## Subject Code: BCACC203

## Subject Name: Computer Architecture

Full Marks: 30 | Set No: 3 | Duration: 1 Hour

### Part - A

**Attempt 5 questions out of 6**

**Each question carries 2 marks** $(2 \times 5)$

1. What is tag field? 2

2. What is Transfer time? 2

3. Define Bus Request. 2

4. What is efficiency in Pipeline? 2

5. What is use of Location Counter (LC)? 2

6. What is BSA instruction? 2

### Part - B

**Attempt 2 questions out of 3**

**Each question carries 5 marks** $(5 \times 2)$

7. Explain the concept of hit ratio and miss ratio in cache memory. 5

8. Draw and analyse the control timing signal for the operation $D_3T_4: SC \leftarrow 0$ 5

9. Draw and explain the flowchart for interrupt cycle. 5

### Part - C

**Attempt 1 question out of 2**

**Each question carries 10 marks** $(10 \times 1)$

10. Draw and explain the flowchart for Booth's multiplication algorithm with an example. 10

11. Draw and explain the flowchart for second pass assembler. 10